

Data Cleaning:

1. Checking the number of missing values marked with '?' in each column of the "Medical Examinations" Table and "Hospitalization Details" Table.

To count the missing values marked with '?' in Excel, can use the COUNTIF function. This function allows to scan an entire column or a specific range and count only the cells that match your exact text.

=COUNTIF(A2:H2336,"?") in the medical examination

=COUNTIF(A2:I2344,"?") in the hospitalisation

2. Fill in the missing values of 'month' with Sep and 'year' with its average rounded to the nearest integer.
 - GO TO option provides to fill the missing values.
 - Average year is calculated to be 1982.84 which is rounded to 1983 by using the average() of the Year column.Missing values in the year is filled with IF function with the formula:
=IF(B2="?",1983,B2)
3. Determine the most frequently occurring values in the 'smoker', 'Hospital tier' and 'City tier' columns, and fill in the missing values accordingly.
 - Missing values are imputed using the IF condition and using PROPER function to further standardise.
=IF(H2="?", "No", PROPER(H2)) for smoker column
=IF(G2="?", "tier - 2", G2) for Hospital tier
=IF(H2="?", "tier - 2", H2) for City tier
4. If any 'State ID' values are missing, consider filling them with 'Unknown' or using another appropriate strategy.
=IF(I8="?", "Unknown", I8)

Data Transformation:

1. Split the 'names' column in the "Customer Names" Table into 3 meaningful columns: 'Title', 'First Name', and 'Last Name'.
Select name column → Data → Text to Columns → Delimited → Comma
Then split the remaining name portion by spaces.
2. Convert the "NumberOfMajorSurgeries" column in the "Medical Examinations" Table to numerical data by replacing non-numeric characters with meaningful numerical values.
=IF(G2="No major surgery",0,G2)
3. Check for inconsistencies in the 'Heart Issues' and 'smoker' columns
It is corrected using the PROPER()
4. Create a new column named "Weight Status" that categorizes BMI into different categories
 - To categorise into Underweight, Normal weight and Overweight

=IF(B2<18.5,"Underweight",IF(B2<=24.9,"NormalWeight",IF(B2<=29.9,"Overweight",IF(B2>30,"Obesity")))))

5. Diabetes Status

- To categorise into Normal, Prediabetes and Diabetes
=IF(C2<5.7,"Normal",IF(C2<=6.4,"Prediabetes ",IF(C2>6.4,"Diabetes ")))

6. Merge 'year', 'month' and 'date' columns in the "Hospitalization Details" Table into one column named 'Date of Birth' and format it in 'DD-MMM-YYYY' custom format.

=CONCAT(D2,"-",C2,"-",B2) and format using the custom option.

7. Calculate the 'Age' of each customer based on their 'Date of Birth' and the date of collection of the dataset, which is 8th June 2023.

=DATEDIF(N2,DATE(2023,6,8),"Y")

8. Format 'charges' column as currency (\$)

Select the charges column and format using the number format by changing into currency.

Data Exploration, Analysis & Visualization:

CREATE HEALTHCARE SHEET

- Create a new worksheet named Healthcare.
- combine all three tables using **Customer ID and VLOOKUP**
- The other columns would include Customer ID, First Name, BMI, HbA1C, Heart Issues, Any Transplants, Cancer history, NumberOfMajorSurgeries, smoker, Weight Status, Diabetes Status, Date of Birth, charges, Hospital tier, City tier, State ID, Age.
- The Customer remains the lookup value for all columns.

PIVOT TABLE ANALYSIS

Analysis 1 — Cancer History among Smokers vs Non-Smokers

For PivotTable

Insert → PivotTable

Source: Healthcare

For inserting Chart, Select PivotTable

Insert → Pie/Donut Chart

Analysis 2 — Transplants vs Surgeries and HbA1C

Create another PivotTable where rows would be Any Transplants and values would include number of surgeries and HbA1c. Also inserting chart accordingly.

Analysis 3 — Charges by Weight Status

Create PivotTable having rows as Weight status and values as charges. Insert Column chart displaying different categories.

Charges by Diabetes Status

Create PivotTable having Row as diabetes status and values as charges.

Analysis 4 — Average Charges by Hospital Tier within States

Create a PivotTable where rows would be State ID, Columns Hospital tier and Values as charges.

Analysis 5 — Age vs BMI and Hba1c

Select both the columns and insert the required chart

Analysis 6 — Age vs charges

Select both the columns and insert the required chart

Dashboard creation

Add the Weight Status and Diabetes Status slicers and the visualisation charts created using pivot to the dashboard.